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Sensor stability review

Profile report with descriptive time-series statistics, generated 2026-10-18 17:39 中国标准时间 by `scripts/assemble_profile_report.py` from the artifacts listed in `report_build.json`. The statistics below are descriptive and screening only -- no forecast, no causal claim.

Executive summary

The input was routed to 1 branch and profiled without changing the raw files. The profile found **3 anomaly flag(s)**, including **2 at P0/P1**. Descriptive time-series statistics were run on `temperature_c` (n=102): no detectable monotone trend; Sen slope -0.00142061 per step (95% CI [-0.0049697, 0.00238095]). This report describes those findings and stops before forecasting or causal inference because the selected branch does not implement a defensible statistical model for this modality; the statistics above are descriptive, not predictive.

1. Scope and routing

The router recorded the following evidence:

Modality	Files	Profiler	Next stage
`time-series`	1	`scripts/profile_timeseries.py`	`cleaning_plan.py`

Profile-only mode inspects and reports. It does not clean, impute, deduplicate, transform, test or draw inferential conclusions. The time-series statistics in section 2 are an explicitly descriptive exception: trend, autocorrelation, decomposition strength and a changepoint screen.

2. Profile findings

2.1 time-series

Field	Value
Modality	`time-series`
Source	`sensor_log.csv`
Records inspected	103
Fields	4
Independent-unit hint	one row per timestamp (heuristic, not a declaration)
Anomaly flags	3
Highest severity	P1

Time-series structure

Property	Observed
Time column	`timestamp`
Modal interval	1 hour
Regular sampling	no
Detected gaps	14
Regime change suspected	yes
First timestamp	2026-01-01T00:00:00
Last timestamp	2026-01-05T02:00:00

Numeric series

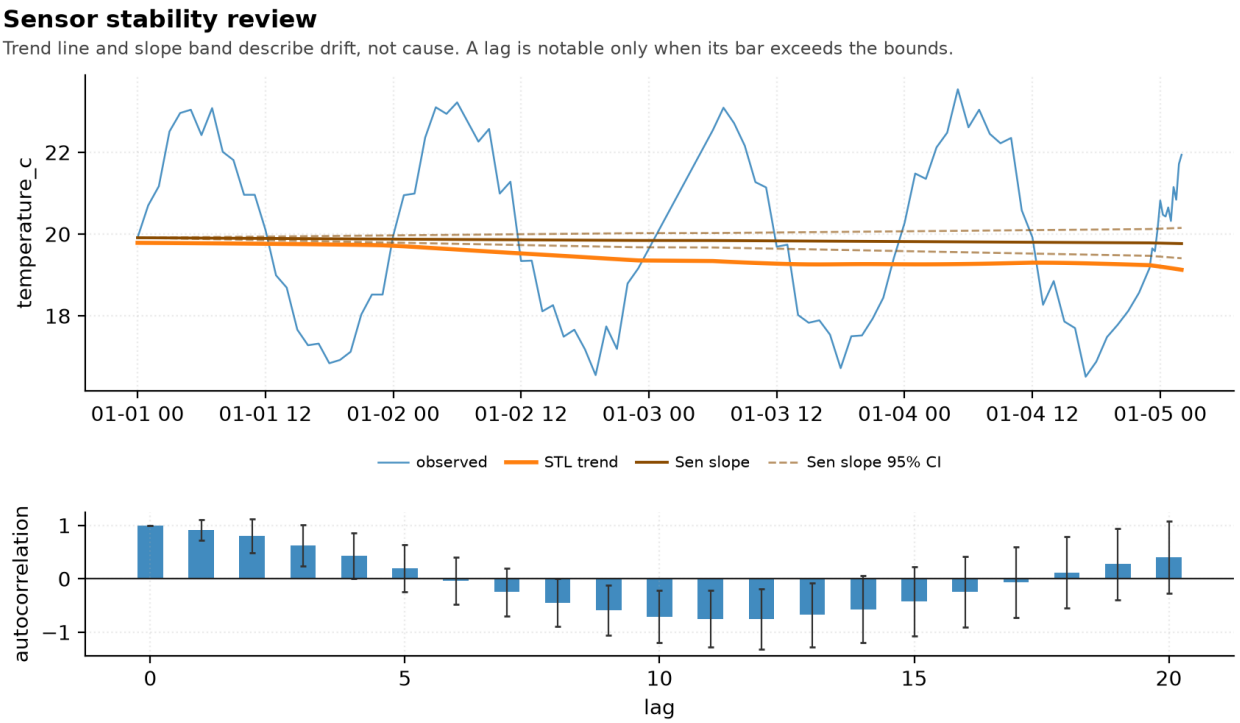
Column	n	Missing	Mean	SD	Min	Max
temperature_c	103	0	19.99	2.049	16.51	23.54
humidity_pct	103	0	54.91	1.897	49.6	58.4

2.2 Descriptive time-series statistics

Statistic	Method	Result		
mann_kendall	Mann-Kendall trend test (two-sided, tie-corrected)	S = -244, p = 0.4822954397073186 -- no detectable monotone trend		
sen_slope	Theil-Sen slope, 95% CI	-0.00142061 per step (CI [-0.0049697, 0.00238095]); -30.68517139 per day		
stl	STL decomposition (robust, period=24)	trend strength 0.01202, seasonal strength 0.75704		
changepoint	CUSUM screen on detrended residuals (permutation null, 500 draws, seed=42)	max	CUSUM	= 11.87487, p = 0.09181636726546906 -- no changepoint detected at alpha=0.05

Every p-value above is two-sided at alpha=0.05 with its test name and n recorded in `analysis/timeseries.json`. None of them licenses a forecast or a causal statement.

Autocorrelation: lags exceeding the white-noise bounds are **none** (ACF) and **none** (PACF), out of 20 examined.



3. Anomalies

Branch	Severity	Type	Column	Count	Meaning
time-series	P1	duplicate_timestamps	--	1	1 timestamp(s) appear more than once; a mean per instant is not the same as an independent observation per row

Branch	Severity	Type	Column	Count	Meaning
`time-series`	`P2`	`irregular_gaps`	--	14	the sampling interval is not constant; a mean over rows is then a mean over sampled instants, not over time
`time-series`	`P1`	`sampling_regime_change`	--	--	the median interval in the last third differs from the first third by more than 3x; check whether the instrument changed

4. What was and was not done

Completed. Format routing, bounded decoding, structural profiling, missingness counts, duplicate checks and modality-specific integrity flags. Descriptive time-series statistics (trend, slope with CI, autocorrelation with white-noise bounds, STL strengths, a seeded changepoint screen) with every p-value carrying its test name and n.

Not performed. Forecasting and ARIMA-style model fitting on the time series. Statistical inference, causal interpretation, automatic row deletion, imputation, deduplication, feature engineering and model fitting. Their absence is a scope boundary, not a passing result.

5. Reproducibility

The run used Python 3.13.15 on Windows-11-10.0.26200-SP0. The profile and route JSON are the evidence for every statement in this report; their hashes and the PDF backend trail are stored in `report_build.json`.

6. Open questions

1. **AUTHOR_INPUT_NEEDED: what does one independent unit represent?** The profiler's unit hint is a heuristic based on the input shape, not a study-design declaration.
2. **AUTHOR_INPUT_NEEDED: which anomalies are genuine defects?** Duplicate documents, near-duplicate images, repeated timestamps and coincident points can be either technical repeats or independent observations; the data alone cannot decide.
3. **AUTHOR_INPUT_NEEDED: what comparison or prediction question is the report meant to answer?** A profile can support exploratory decisions but cannot choose a model without a stated estimand.

End of profile-only report. No value in the raw input was modified.